

XT MCU Android API Document

Version	Modified by	Date	Description
V1.0.0	Kelly	2023-05-25	First edition
V1.1.0	Kelly	2023-07-27	Add to gain the mcu dashboard information
V1.2.0	Kelly	2023-08-24	Front light control
V1.3.0	Kelly	2023-10-19	Increase the low-power turn-on command
V1.4.0	Kelly	2023-11-23	Add the MCU OTA command
V1.4.1	Kelly	2024-01-11	Perfect the MCU OTA
V1.4.2	Kelly	2024-04-16	Battery status reporting and obtaining control board information support returned power percentage
V1.5.0	Kelly	2024-05-30	Increase the front brightness adjustment command
V1.5.1	Kelly	2024-10-22	Optimize serial port communication for stronger compatibility
V1.6.0	Kelly	2024-12-16	Add the power board LED lamp control
V1.6.1	Kelly	2025-01-16	Improve abnormal handling, improve stability

Catalog

Catalog.....	1
1 Introduction.....	2
1.1 Purpose.....	2
1.2 Document range.....	2
1.3 Expect readers.....	2
2 Android Studio Import.....	2
3 Description of the API interface function.....	4
3.1 Open.....	4
3.2 Close.....	4
3.3 Gets the connection status.....	4
3.4 Low power on.....	4
3.5 Send the heartbeat.....	4
3.6 Power supply control.....	5
3.7 Adjust the front light brightness.....	5
3.8 Power board LED lamp control.....	5
3.9 Time-lapse boot.....	6
3.10 Battery status listening.....	6
3.11 Mcu key listening.....	7
3.12 Get the Mcu control board information.....	7
3.13 Mcu OTA upgrade.....	8
4 Status Code.....	8

1 Introduction

1.1 Purpose

This document describes the API library [xt-mcu-1.6.1.jar](#) of Android and mcu communication, realizing the low power consumption function based on serial port communication.

1.2 Document range

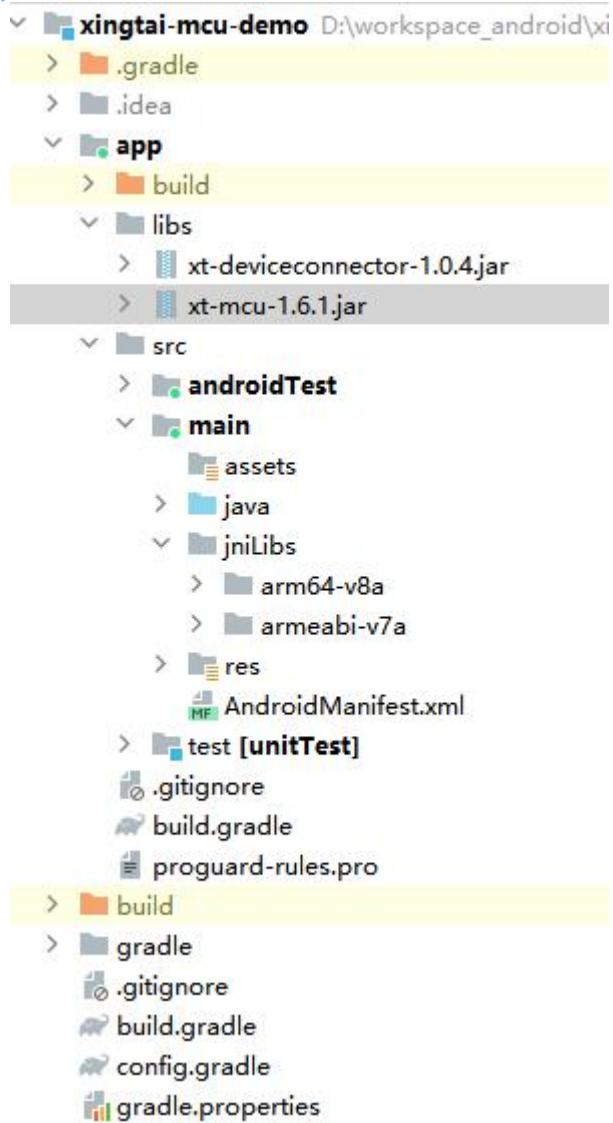
- ✓ [xt-mcu-1.6.1.jar](#)引入
- ✓ API definition
- ✓ API instruction
- ✓ API input and output parameters

1.3 Expect readers

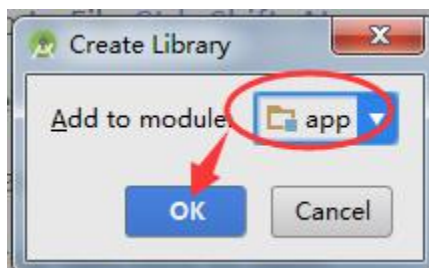
Development, testing, and related approvals

2 Android Studio Import

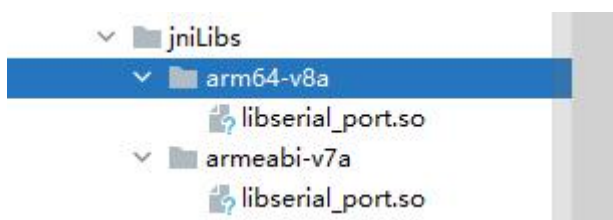
1. Copy the provided [xt-mcu-1.6.1.jar](#) file to the app-> libs folder.
2. Right-click on the added jar file and select Add As Library。



3. Select the corresponding item [module](#), and then click OK.



4. Copy demand copy so library to src/main/jniLibs.



3 Description of the API interface function

3.1 Open

Function: public int open()

Instruction: Open the serial port connection

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.2 Close

Function: public void close()

Instruction: Close the serial port connection

Return value type	Returned value specification	Examples
void	No	

3.3 Gets the connection status

Function: public int getState()

Instruction: Gets the serial port connection status

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.4 Low power on

函数: int startLowPower(boolean enable)

说明: Low power switch. When low power consumption is turned on, if the power board does not receive the user heartbeat for three minutes, when low power consumption is closed, no operation will be done

Input parameter type	Introduction	Example
boolean	enable, true open,false close	

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

Note: This instruction is supported in version 1.0.8 and above

3.5 Send the heartbeat

Function: public int sendHeartbeat()

Instruction: Android Send information to mcu, control by the user to send the heartbeat regularly, time customization, recommended 1 minute once

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.6 Power supply control

Function: public int powerControl(boolean mainPower, int delayTime, Boolean t1000Power, Boolean fivePower)

Instruction: Power supply control, controlled by the service layer itself

Input parameter type	Introduction	Example
boolean	mainPower, Main power off, true no operation, false execution	
int	delayTime, Main power off before delay, 1-255 seconds	
boolean	t1000Power, T1000 power supply, true power is on, false power is off, empty transmission is not operate	
boolean	fivePower, 5V power supply, true power supply is on, false power supply is off, empty transmission does not operate	It can be used here for the front optical power supply control

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.7 Adjust the front light brightness

Function: public int adjustBrightness(int brightnessLevel)

Instruction: Adjusting brightness

Input parameter type	Introduction	Example
brightnessLevel	Brightness level, 1~10, subject to the actual equipment	

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

Note: This directive is supported in versions 1.2.0 and above

3.8 Power board LED lamp control

Function: public int ledControl(boolean open, int ledType, int interval, int times)

Instruction: The control LED lights flashes

Input parameter type	Introduction	Example
open	Switch, true on, false off	
ledType	LED light type, 2 is green light, 3 is red light, 4 is blue light	
interval	Flash interval, 0 is constant bright, 1 is interval 1s flicker, unit s, 0~127s	1
times	The number of flashing	1

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

Note: This directive is supported in versions 1.2.5 and above

3.9 Time-lapse boot

Function: `public int boot(int no, int delayTime)`

Instruction: After executing this function, call the power supply control to turn off the total power supply

Input parameter type	Introduction	Example
int	no, Time-lapse boot number	
int	delayTime, Delayed startup time (seconds)	

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.10 Battery status listening

Function: `public void setBatteryChangeListener(BatteryChangeListener batteryChangeListener)`

Instruction: Monitor the battery charging status, power change

Input parameter type	Introduction	Example
BatteryChangeListener	batteryChangeListener, The listening interface The callback function: <code>public void onChange(int charge, int mV, int soc)</code>	

Return value type	Returned value specification	Examples
-------------------	------------------------------	----------

void		
------	--	--

Note: Power percentage soc is supported on version 1.1.9 and above

3.11 Mcu key listening

Function: public void setMcuKeyListener(McuKeyListener mcuKeyListener)

Instruction: Short and long press for the mcu button

Input parameter type	Introduction	Example
McuKeyListener	mcuKeyListener, Listening interface The callback function:public void onKeyDown(int keyCode, int keyDelayFlag, int det)	

Return value type	Returned value specification	Examples
void		

3.12 Get the Mcu control board information

Function: int getMcuInfo(McuInfo mcuInfo);

Instruction: Get the version information of mcu, charging status, and battery voltage

Input parameter type	Introduction	Example
McuInfo	mcuInfo McuInfo { private String version; private boolean hasBattery; private int charge; private int voltage; private int soc; private int brightnessLevel; private String internalVersion; private String chipVersion; }	

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

Note: This directive is supported in versions 1.0.6 and above
, power percentage soc is supported on version 1.1.9 and above

3.13 Mcu OTA upgrade

Function: public int upgradeFirmware(File file)

Instruction: Upgrade the power board firmware

Input parameter type	Introduction	Example
File	file, firmware file, the file name suffix is the bin	

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

Note: This directive is supported in versions 1.1.1 and above

4 Status Code

Description: The actual status code returned shall prevail. The following is the common status code

Status Code	Instruction
0	Success
-1	Communication timeout
-2	Failed to send data (write failure)
-3	Failure to receive data (read failure)
-4	Data validation failed
-5	The device is not connected
-6	The device is connected
-7	Data returned to empty
-8	Operation cancelled
-9	This feature is not supported
-10	Uninitialized
-11	The device was not found
-12	Device driver is not found
-14	No permission
-15	Permission rejected
-16	Open the failure
-99	Communication failure